

Name: Fatima Rza

ERP: 29998

Task 1 and 2

```
hart 1 starting
hart 2 starting
page table 0x0000000087f52000
..0: pte 0x0000000021fd3801 pa 0x0000000087f4e000
.. ..0: pte 0x0000000021fd3401 pa 0x0000000087f4d000
.. ..0: pte 0x0000000021fd3c1b pa 0x0000000087f4f000
.. ..1: pte 0x0000000021fd3017 pa 0x0000000087f4c000
.. ..2: pte 0x0000000021fd2c07 pa 0x0000000087f4b000
.. ..3: pte 0x0000000021fd2817 pa 0x0000000087f4a000
..255: pte 0x0000000021fd4401 pa 0x0000000087f51000
.. ..511: pte 0x0000000021fd4001 pa 0x0000000087f50000
.. ..510: pte 0x0000000021fd5807 pa 0x0000000087f56000
.. ..511: pte 0x000000002000180b pa 0x0000000080006000
init: starting sh
$ pgtbltest
print_pgtbl starting
va 0x0 pte 0x21FC885B pa 0x87F22000 perm 0x5B
va 0x1000 pte 0x21FC7C1B pa 0x87F1F000 perm 0x1B
va 0x2000 pte 0x21FC7817 pa 0x87F1E000 perm 0x17
va 0x3000 pte 0x21FC7407 pa 0x87F1D000 perm 0x7
va 0x4000 pte 0x21FC70D7 pa 0x87F1C000 perm 0xD7
va 0x5000 pte 0x0 pa 0x0 perm 0x0
va 0x6000 pte 0x0 pa 0x0 perm 0x0
va 0x7000 pte 0x0 pa 0x0 perm 0x0
va 0x8000 pte 0x0 pa 0x0 perm 0x0
va 0x9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF6000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF7000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF8000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFA000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFB000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFC000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFD000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFE000 pte 0x21FD08C7 pa 0x87F42000 perm 0xC7
va 0x3FFFFFFF000 pte 0x2000184B pa 0x80006000 perm 0x4B
print_pgtbl: OK
umprint test starting
```

```

print_pgtbl starting
va 0x0 pte 0x21FC885B pa 0x87F22000 perm 0x5B
va 0x1000 pte 0x21FC7C1B pa 0x87F1F000 perm 0x1B
va 0x2000 pte 0x21FC7817 pa 0x87F1E000 perm 0x17
va 0x3000 pte 0x21FC7407 pa 0x87F1D000 perm 0x7
va 0x4000 pte 0x21FC70D7 pa 0x87F1C000 perm 0xD7
va 0x5000 pte 0x0 pa 0x0 perm 0x0
va 0x6000 pte 0x0 pa 0x0 perm 0x0
va 0x7000 pte 0x0 pa 0x0 perm 0x0
va 0x8000 pte 0x0 pa 0x0 perm 0x0
va 0x9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF6000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF7000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF8000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFA000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFB000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFC000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFD000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFE000 pte 0x21FD08C7 pa 0x87F42000 perm 0xC7
va 0x3FFFFFFF000 pte 0x2000184B pa 0x80006000 perm 0x4B
print_pgtbl: OK
vmprint_test starting
page table 0x0000000087f25000
..0: pte 0x0000000021fc8401 pa 0x0000000087f21000
.. ..0: pte 0x0000000021fc8001 pa 0x0000000087f20000
.. .. ..0: pte 0x0000000021fc885b pa 0x0000000087f22000
.. .. ..1: pte 0x0000000021fc7c1b pa 0x0000000087f1f000
.. .. ..2: pte 0x0000000021fc7817 pa 0x0000000087f1e000
.. .. ..3: pte 0x0000000021fc7407 pa 0x0000000087f1d000
.. .. ..4: pte 0x0000000021fc70d7 pa 0x0000000087f1c000
..255: pte 0x0000000021fc9001 pa 0x0000000087f24000
.. ..511: pte 0x0000000021fc8c01 pa 0x0000000087f23000
.. .. ..510: pte 0x0000000021fd08c7 pa 0x0000000087f42000
.. .. ..511: pte 0x000000002000184b pa 0x0000000080006000
vmprint_test: done
pgaccess_test starting
pgtbltest: pgaccess_test failed: incorrect access bits set, pid=3

```

Task 3

```

print_pgtbl: OK
vmprint_test starting
page table 0x0000000087f22000
..0x0000000000000000: pte 0x0000000021fc7801 pa 0x0000000087f1e000
.. ..0x0000000000000000: pte 0x0000000021fc7401 pa 0x0000000087f1d000
.. .. ..0x0000000000000000: pte 0x0000000021fc7c5b pa 0x0000000087f1f000 RXU
.. .. ..0x00000000000001000: pte 0x0000000021fc701b pa 0x0000000087f1c000 RXU
.. .. ..0x00000000000002000: pte 0x0000000021fc6c17 pa 0x0000000087f1b000 RWU
.. .. ..0x00000000000003000: pte 0x0000000021fc6807 pa 0x0000000087f1a000 RW
.. .. ..0x00000000000004000: pte 0x0000000021fc64d7 pa 0x0000000087f19000 RWU
..0x00000003fc000000: pte 0x0000000021fc8401 pa 0x0000000087f21000
< ..0x00000003fffe0000: pte 0x0000000021fc8001 pa 0x0000000087f20000
.. .. ..0x00000003fffffd000: pte 0x0000000021fd4813 pa 0x0000000087f52000 RU
.. .. ..0x00000003fffffe000: pte 0x0000000021fd00c7 pa 0x0000000087f40000 RW
.. .. ..0x00000003ffffff000: pte 0x0000000020001c4b pa 0x0000000080007000 RX
vmprint_test: done
pgaccess_test starting
pgaccess_test: OK
ugetpid_test starting
ugetpid_test: OK
pgtbltest: ksuper_test failed: not enough of super ptes, pid=3

```

Task 4 and 5

```
$ pgtbltest
print_ptbl starting
va 0x0 pte 0x21FDF85B pa 0x87F7E000 perm 0x5B
va 0x1000 pte 0x21FDEC1B pa 0x87F7B000 perm 0x1B
va 0x2000 pte 0x21FDE817 pa 0x87F7A000 perm 0x17
va 0x3000 pte 0x21FDE407 pa 0x87F79000 perm 0x7
va 0x4000 pte 0x21FDE0D7 pa 0x87F78000 perm 0xD7
va 0x5000 pte 0x0 pa 0x0 perm 0x0
va 0x6000 pte 0x0 pa 0x0 perm 0x0
va 0x7000 pte 0x0 pa 0x0 perm 0x0
va 0x8000 pte 0x0 pa 0x0 perm 0x0
va 0x9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF6000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF7000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF8000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFA000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFB000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFC000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFD000 pte 0x21FEC413 pa 0x87FB1000 perm 0x13
va 0x3FFFFFE000 pte 0x21FE7CC7 pa 0x87F9F000 perm 0xC7
va 0x3FFFFF000 pte 0x20001C4B pa 0x80007000 perm 0x4B
print_ptbl: OK
vmprint_test starting

vmprint_test starting
page table 0x0000000087f81000
..0x0000000000000000: pte 0x0000000021fdf401 pa 0x0000000087f7d000
.. ..0x0000000000000000: pte 0x0000000021fdf001 pa 0x0000000087f7c000
.. .. ..0x0000000000000000: pte 0x0000000021fdf85b pa 0x0000000087f7e000 RXU
.. .. ..0x00000000000001000: pte 0x0000000021fdec1b pa 0x0000000087f7b000 RXU
.. .. ..0x00000000000002000: pte 0x0000000021fde817 pa 0x0000000087f7a000 RWU
.. .. ..0x00000000000003000: pte 0x0000000021fde407 pa 0x0000000087f79000 RW
.. .. ..0x00000000000004000: pte 0x0000000021fde0d7 pa 0x0000000087f78000 RWU
..0x00000003fc000000: pte 0x0000000021fe0001 pa 0x0000000087f80000
.. ..0x00000003fffe0000: pte 0x0000000021fdcf01 pa 0x0000000087f7f000
.. .. ..0x00000003ffffd000: pte 0x0000000021fec413 pa 0x0000000087fb1000 RU
.. .. ..0x00000003ffffe000: pte 0x0000000021fe7cc7 pa 0x0000000087f9f000 RW
.. .. ..0x00000003fffff000: pte 0x0000000020001c4b pa 0x0000000080007000 RX
vmprint_test: done
pgaccess_test starting
pgaccess_test: OK
ugetpid_test starting
ugetpid_test: OK
ksuper_test: OK
pgtbltest: all tests succeeded
```

Grading Test

```
== Test pgtbltest ==  
$ make qemu-gdb  
(7.2s)  
== Test  pgtbltest: vmprint_test ==  
pgtbltest: vmprint_test: OK  
== Test  pgtbltest: pgaccess ==  
pgtbltest: pgaccess: OK  
== Test  pgtbltest: ugetpid ==  
pgtbltest: ugetpid: OK  
== Test  pgtbltest: superpg ==  
pgtbltest: superpg: OK  
== Test answers-pgtbl.txt ==  
answers-pgtbl.txt: OK  
== Test usertests ==  
$ make qemu-gdb  
(153.0s)  
== Test time ==  
time: OK  
Score: 46/46  
fatima@fatima-VMware-Virtual-Platform:~/xv6-riscv$
```